

Week 4 Lab: Data sources, collection and sampling

INF2167 - R for Data Science

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In this lab, you'll work with a survey dataset and explore how data is constructed through decisions about who is included, how responses are measured, and how categories are defined. You will also reflect on how these choices shape what we can and cannot learn from data.

Before you begin, update the author name in the document front matter to your first and last name. Then, complete the lab by writing your code and answers directly below each question. Submit your .qmd file and PDF on Quercus. You may work individually or in small groups. For the group project section, you and your partner should work together on ideas, but each member of the group should submit their own files on Quercus.

Individual/Group: Working with survey data

Set up and inspecting our dataset

First, let's set up our workspace and read in the cleaned Canadian Election Study (CES) dataset. I've selected a few variables from the 2019 CES (web) and done some preliminary cleaning. If you are interested in learning more about the 2019 CES, check out the article: [*Measuring preferences and behaviours in the 2019 Canadian Election Study*](#)

```
## Load package ##  
library(tidyverse)  
  
## Read in the dataset ##  
ces_data <- read_csv("data/cleaned_ces.csv", show_col_types = FALSE)
```

Now, let's get an overview of the dataset:

```
glimpse(ces_data)
```

```
Rows: 14,833
Columns: 8
$ ID      <chr> "R_10pYXEFGzHRUpjM", "R_2qdrL3J618rxYW0", "R_2EcS26hqrcVYla~
$ Age     <dbl> 30, 21, 21, 19, 21, 21, 19, 19, 20, 19, 19, 22, 20, 19, 19,~
$ Gender  <chr> "A woman", "A woman", "A woman", "A woman", "A man", "A wom~
$ Province <chr> "Quebec", "Quebec", "Ontario", "Ontario", "Ontario", "Ontar~
$ Education <chr> "Master's degree", "Master's degree", "Some post-secondary"~
$ Religion <chr> "Don't know/ Prefer not to answer", "Catholic", "Catholic",~
$ Employment <chr> "Don't know/ Prefer not to answer", "Student and working", ~
$ Ideology <dbl> 2, 2, 2, 4, 2, 1, 6, 6, 1, 4, 1, 10, 9, 2, 4, 2, 3, 2, 5, 8~
```

What do you notice about the values in the **Gender** variable? Are the categories clear and easy to read? Or are they overly long and detailed?

Re-coding

Now, we will begin changing how variables are represented in the dataset. Remember from the lecture that this is called re-coding, and it is one way that we can transform raw data into cleaner, analyzable data.

First, look at the current values of the **Gender** variable.

```
unique(ces_data$Gender)
```

```
[1] "A woman"
[2] "A man"
[3] "Other (e.g. Trans, non-binary, two-spirit, gender-queer)"
```

You should see categories like “A woman” and “A man”. These are somewhat clear, but could be clearer for analysis.

Let’s re-code the **Gender** variable options using the code we went over in class. After re-coding, “A woman” should be “Woman”, “A man” should be “Man”, and “Other (e.g. Trans, non-binary, two-spirit, gender-queer)” should be “Other”.

Write your code in the code chunk below:

```
# YOUR CODE HERE
```

Exploring the dataset

Now that we have cleaned and re-coded some variables, we will begin exploring what the dataset looks like. Remember from the lecture that a sample is only a subset of a population and different sampling processes can shape what we observe.

Province distribution

Let's start by looking at where respondents are from:

```
ces_data |>
  count(Province) |>
  arrange(desc(n))
```

```
# A tibble: 13 x 2
  Province          n
  <chr>          <int>
1 Ontario        5854
2 Quebec         3176
3 Alberta        1854
4 British Columbia 1804
5 Manitoba        629
6 Saskatchewan    512
7 Nova Scotia     354
8 New Brunswick   325
9 Newfoundland and Labrador 227
10 Prince Edward Island 59
11 Yukon          22
12 Nunavut         9
13 Northwest Territories 8
```

What do you notice about the distribution of respondents across provinces? Are some provinces more represented than others?

Respondent age

Now, let's look at the age of respondents:

```
ces_data |>
  summarize(
    min_age = min(Age, na.rm = TRUE),
    max_age = max(Age, na.rm = TRUE),
    mean_age = mean(Age, na.rm = TRUE),
    median_age = median(Age, na.rm = TRUE))
```

```
# A tibble: 1 x 4
  min_age max_age mean_age median_age
  <dbl>   <dbl>   <dbl>     <dbl>
1     18     99    48.5       49
```

What is the typical age range in this sample? Are there any potential limitations in how age is distributed?

Respondent education

Looking at the Education variable now, which levels of education attained are the most and least represented in the dataset?

```
ces_data |>
  group_by(Education) |>
  count() |>
  ungroup() |>
  mutate(percentage = round(n / sum(n) * 100, 2)) |>
  arrange(desc(n))
```

```
# A tibble: 11 x 3
  Education                n percentage
  <chr>                <int>     <dbl>
1 Bachelor's degree    3959     26.7
2 Some post-secondary  3072     20.7
3 College/technical diploma 2843     19.2
4 High school          2085     14.1
5 Master's degree      1448      9.76
6 Professional degree or doctorate 664      4.48
```

7	Some high school	570	3.84
8	Elementary school	112	0.76
9	Some elementary school	38	0.26
10	Don't know/ Prefer not to answer	25	0.17
11	No school	17	0.11

Respondent employment

Now, let's look at the Employment variable.

```
ces_data |>
  group_by(Employment) |>
  count() |>
  ungroup() |>
  mutate(percentage = round(n / sum(n) * 100, 2)) |>
  arrange(desc(n))
```

A tibble: 13 x 3

	Employment	n	percentage
	<chr>	<int>	<dbl>
1	Working full-time	6406	43.2
2	Retired	3531	23.8
3	Working part-time	1336	9.01
4	Self-employed	1042	7.02
5	Unemployed	586	3.95
6	Student	537	3.62
7	Disabled	481	3.24
8	Caring for family	394	2.66
9	Other	145	0.98
10	Retired and working	127	0.86
11	Don't know/ Prefer not to answer	113	0.76
12	Student and working	83	0.56
13	Caring for family and working	52	0.35

What do you notice about how employment status is coded in this dataset? Are people always in a single clear category or do some categories overlap?

Ideology

Now we will examine a variable that is often used to measure political affiliation: **Ideology**. This is often measured on a left-right scale where respondents place themselves.

Using `unique()` shows all the different values that appear in the `Ideology` variable, but it does not tell us how those values are distributed or what they represent.

```
unique(ces_data$Ideology)
```

```
[1]  2  4  1  6 10  9  3  5  8  0  7
```

Now, go to the [2019 Canadian Election Study Codebook](#). Search for the variable `cps19_lr_scale_aft_1` (I re-named it to `Ideology`). Using the codebook, answer the following questions:

- What do the lowest and highest values on the scale represent?
- What does a value in the middle of the scale represent?
- How does self-placement potentially shape the data collected?

Sampling variability

So far, we have been working with the full dataset. In real research, however, we often work with samples, not full populations. To see how sampling affects results, we will draw two different random samples from the same dataset and compare them.

First random sample

We set a seed using `set.seed()` so that results are reproducible, then draw a random sample of 200 respondents using the `slice_sample()` function:

```
set.seed(16)

sample1 <- ces_data |>
  slice_sample(n = 200)
```

Now calculate the mean age in the full dataset and in this sample:

```
## Full dataset ##
mean(ces_data$Age, na.rm = TRUE)
```

```
[1] 48.54365
```

```
## Sample ##  
mean(sample1$Age, na.rm = TRUE)
```

```
[1] 48.945
```

Second random sample

Now we repeat the process with a different seed:

```
set.seed(26)  
  
sample2 <- ces_data |>  
  slice_sample(n = 200)
```

Compare again:

```
## Full dataset ##  
mean(ces_data$Age, na.rm = TRUE)
```

```
[1] 48.54365
```

```
## Sample ##  
mean(sample2$Age, na.rm = TRUE)
```

```
[1] 47.23
```

Questions

- Are the sample means identical to the full dataset mean? Why or why not?
- How do sample1 and sample2 differ from each other?
- What does this tell us about using samples to make claims about a population?

Sampling and group composition

Now, let's look at whether the composition of categories also changes across samples:

```
sample1 |> count(Gender)
```

```
# A tibble: 2 x 2
  Gender      n
  <chr>   <int>
1 A man     79
2 A woman  121
```

```
sample2 |> count(Gender)
```

```
# A tibble: 3 x 2
  Gender      n
  <chr>   <int>
1 A man     83
2 A woman  114
3 Other (e.g. Trans, non-binary, two-spirit, gender-queer)  3
```

Questions

- Do the counts for Gender look the same in both samples? How do they differ?
- What does this suggest about using a single sample to describe a population?

Bonus: Graphing

Applying some of the data visualization techniques we learned last week (week 3), create at least one plot exploring a relationship between two variables in the dataset. It is up to you which variables you select and what type of graph you choose to make.

Some questions you may consider answering include:

- Does ideology vary by gender or education?
- How is age distributed across provinces?
- Are there visible differences in education levels across groups?

```
# GRAPH HERE
```


Group project: Dataset construction and analysis

Based on the types of datasets we discussed in today's lecture and some of the others that we will discussing in more detail during week 6, what type of dataset do you think you'll construct and use for your project proposal and paper?

Who might be missing from your dataset and how could you account for that in your analysis?

How does the way your key concept is measured affect what you can learn from your dataset? What might this measure leave out or simplify too much?